MIS: A Difficulty Concentration Signal for Pre Copy-Paste Augmentation Diagnosis in Agricultural Object Detection

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***Abstract*—Copy-paste augmentation achieves strong results for instance segmentation yet yields inconsistent or negative gains for bounding-box detection in agricultural scenes. We propose the Minimum Inference Score (MIS), a difficulty metric derived from a trained baseline detector that correlates with detection failure modes—miss rate and false-positive rate—across objects and images. MIS reveals where a model fails in a dataset: when MIS detects strong or moderate concentration of hard instances in near-zero AP regimes—regardless of whether the structural source is object size, domain composition, or both—copy-paste augmentation cannot provide reliable gains. We demonstrate this as a pre-augmentation diagnostic validated across three agricultural benchmarks, where MIS correctly anticipates failure under structural constraints and consistent gains in their absence.**

***Index Terms*—object detection, copy-paste augmentation, difficulty scoring, agricultural object detection, data augmentation diagnostic**

I. Introduction

Copy-paste augmentation composites cropped labeled instances into training images to increase sampling diversity [1], [2]. While effective for instance segmentation [2], bounding-box detection in agricultural imagery is subject to harsher constraints—sub-resolution instances, multi-domain acquisition, and high object density—frequently producing inconsistent or negative augmentation outcomes and forcing costly multi-seed search without a principled stopping criterion.

We introduce the **Minimum Inference Score (MIS)**, a metric computed from a trained baseline detector that captures model failure patterns on a given dataset. By correlating with miss rate and FP rate, MIS surfaces where difficulty concentrates in a dataset (driven by object size, domain composition or their interaction) and helps determines whether copy-paste can provide useful training signal. On MinneApple [3], GWHD 2021 [4], and WGISD [5], MIS correctly anticipates failure when hard instances concentrate in structurally constrained regimes and consistent gains when no such constraint is present, providing bidirectional diagnostic validation.

II. Methodology

A. Scoring Formulation

Let $G(I)$ denote ground-truth boxes in image I. A trained detector produces predictions p with confidence c_p at $\tau_{\text{conf}}{=}0.001$ to retain near-miss candidates. For each $g \in G(I)$, let $\mathcal{P}(g){=}\{p : \text{IoU}(p,g){\geq}0.5\}$. The *Object Difficulty Score* is

$$S_{\text{obj}}(g) = \begin{cases} \min_{p\in\mathcal{P}(g)}[\alpha(1{-}c_p) + \beta(1{-}u_p)], & \mathcal{P}(g) \neq \varnothing \\ \alpha+\beta, & \mathcal{P}(g) = \varnothing \end{cases} \tag{1}$$

with $\alpha{=}\beta{=}0.5$ and $u_p{=}\text{IoU}(p,g)$. Unmatched objects receive the maximum score ($\alpha{+}\beta{=}1$). The *Image Difficulty Score* is

$$S_{\text{img}}(I) = \frac{1}{|G(I)|}\sum_g S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I), \tag{2}$$

where $\text{FPrate}(I)$ counts unmatched predictions per image. This yields $S_{\text{obj}} \in [0,1]$ for every ground-truth object and S_{img} for every training image, where higher values indicate greater detection difficulty. By construction, S_{img} encodes both axes of detection failure (miss rate and FP rate); w_2 is set by grid search to balance their contributions.

B. Augmentation Protocol

We train a YOLOv8s baseline and partition all training objects by S_{obj} and all training images by S_{img} into equal difficulty tertiles (Low, Medium, High). Four copy-paste conditions are evaluated: **Score High** pastes objects from the highest-difficulty object tertile into the highest-difficulty destination images; **Score Medium** and **Score Low** follow the same logic for their respective tertiles. **Random** selects both destination images and source objects without score-based filtering. This design isolates whether directing augmentation toward specific difficulty regimes affects outcome. Following [1], training data is expanded by 30% via unblended within-image copy-paste: 2–8 objects per image, scale $[0.9, 1.1]$, rotation $\pm 15°$, $3\times$ maximum reuse; all other settings follow the default YOLOv8 protocol.

III. Results

A. MIS Reflects Detection Failure Modes

Pearson correlations between S_{img} and miss rate are $r{=}0.66$, 0.75, 0.63 on MinneApple, GWHD, and WGISD; correlations with FP rate are $r{=}0.63$, 0.75, 0.64. Consistent moderate-to-strong correlations across all three datasets confirm that MIS captures the primary axes of detection failure as intended by its construction.

TABLE I

MINNEAPPLE: S_{obj} CAPTURES SIZE-LEVEL DIFFICULTY.

Category	AP	Mean S_{obj}	Top tertile
Very Small ($\leq$400 px^2)	0.071	0.152	64.4%
Small (400–1024 px^2)	0.270	0.076	33.9%
Medium ($>$1024 px^2)	0.493	0.053	1.7%

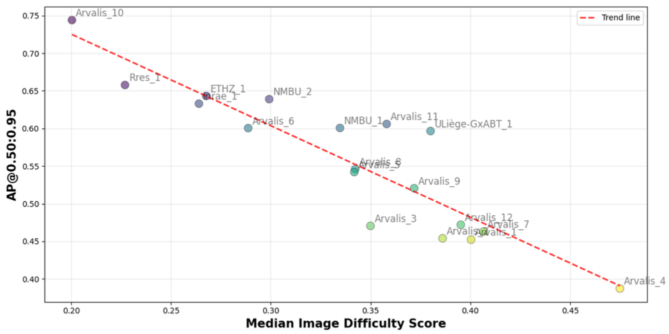

Fig. 1. GWHD 2021: per-domain AP vs. median image difficulty (MIS).

B. MIS Identifies Structural Failure Sources

Size (MinneApple). S_{obj} inversely tracks AP across size groups (Spearman r_s=$-$0.82; Table I): 64.4% of the top-difficulty tertile falls in *Very Small* instances ($\leq$400 px^2), where AP is near-zero (0.071), indicating hard objects concentrate below the model's size recognition threshold.

Domain (GWHD 2021). Per-domain median MIS strongly anticorrelates with domain AP (Pearson r=$-$0.90; Fig. 1). A concentration test confirms that the top 20% hardest images are statistically enriched for near-zero AP objects (1.34$\times$ lift over random), though the effect is moderate, suggesting GWHD difficulty is multi-factorial rather than traceable to a single structural source.

C. Application: Pre-Augmentation Diagnostic for Copy-Paste

MinneApple (resolution ceiling). MIS identifies hard objects concentrated below the recognition threshold. All four copy-paste conditions yield negative AP (Table II) showing that: adding more sub-resolution instances provides no usable learning signal regardless of selection strategy.

GWHD 2021 (domain instability). MIS identifies hard images clustering in specific domains, with difficulty that is multi-factorial rather than traceable to a single structural source. The baseline itself carries notable seed variance (0.229$\pm$0.013), reflecting the instability of multi-domain training. All copy-paste conditions produce negative mean AP. Score Low yields the worst outcome ($-$0.034$\pm$0.009, a relative drop of 14.8%), while Score High achieves the least negative mean ($-$0.006) but extreme seed variance ($\pm$0.040). The moderate near-zero AP concentration (1.34$\times$ lift, p<0.001) and high baseline variance ($\pm$0.013) together indicate that GWHD difficulty is multi-factorial; MIS correctly flags the dataset as high-risk for copy-paste despite the complex underlying structure.

TABLE II

COPY-PASTE OUTCOMES (3-SEED MEAN $\pm\,\sigma$). CONC. = NEAR-ZERO AP ENRICHMENT IN TOP-20% HIGH-MIS IMAGES.

Dataset	Conc.	Baseline AP	Best CP (Δ)	Worst CP (Δ)
MinneApple	Strong	0.353 ± 0.005	-0.003 (Hi)	-0.008 (Rnd)
GWHD 2021	Moderate	0.229 ± 0.013	$-0.006^{\dagger}$ (Hi)	-0.034 (Lo)
WGISD	Absent	0.480 ± 0.007	$+0.009$ (Med)	$+0.001$ (Hi)

†Score High on GWHD: σ=0.040, high seed variance.

WGISD (positive control). MIS flags no structural bottleneck: hard instances do not concentrate below the size threshold or in minority domains. All conditions yield positive gains (+0.001 to +0.009; Table II), with Score Medium achieving the best result (0.489$\pm$0.006 vs. baseline 0.480$\pm$0.007), confirming copy-paste viability when MIS raises no flag.

Across all three benchmarks, the *direction* of copy-paste outcome is consistent with the MIS diagnostic: all four conditions are negative on flagged datasets and positive on the unflagged dataset. This directional consistency, independent of which selection condition is used, supports a structural rather than selection-driven explanation.

IV. CONCLUSION

MIS links baseline detector failure behavior to difficulty concentration patterns, enabling a practical pre-augmentation signal that indicates whether copy-paste is likely to yield reliable gains. In our experiments, 24 augmented training runs across MIS-flagged datasets (MinneApple and GWHD, 4 conditions $\times$ 3 seeds each) yielded zero reliable gains; MIS identifies these datasets without any augmented run, directly saving that compute budget. Practitioners should apply MIS after baseline training: when MIS identifies strong or moderate concentration of high-difficulty instances in near-zero AP regimes, copy-paste augmentation is unlikely to yield reliable gains. This finding is validated bidirectionally across three agricultural benchmarks. Future work includes using MIS to guide instance-level selection when structural constraints are absent, and combining MIS with generative methods to produce information-rich training instances for hard regimes.

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